Combinatorial Geometry

Labix

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1 Simplicies

1.1 Basic Definitions

Definition 1.1.1 (Affinely Independent)
Let $n, m \in \mathbb{N}$. We say that a set of points $\{v_0, \dots, v_m\} \subset \mathbb{R}^n$ is affinely independent if $v_1 - v_0, \dots, v_n - v_0$ are linearly independent.

Definition 1.1.2 (n-Simplexes)
Let $n \in \mathbb{N}$. Let $v_0, \dots, v_n \in \mathbb{R}^n$ be a set of affinely independent points. An n-simplex is the set of points

$$\Delta^n = \left\{ \sum_{k=0}^n t_k v_k \,\middle|\, \sum_{k=0}^n t_k = 1 \text{ and } t_k \geq 0 \text{ for all } k = 0, \dots, n \right\}$$

We write $\Delta^n = [v_0, \dots, v_n]$ to indicate the spanning vectors. The standard n-simplex is just the n-simplex whose vertices are the standard basis vectors for $\mathbb{R}^{n+1}$.

Note that the vertices here are an ordered set so that an orientation is inherently defined. Realistically, the order of the vertices does not change the homology groups which we will define later.

Definition 1.1.3 (Properties of n-Simplexes) Let $\Delta^n = [v_0, \dots, v_n]$ be an n-simplex.

- The kth face of Δ^n is a the $n-1$ simplex $\Delta^n_k = \Delta^n \cap \{x_k = 0\}$. We use $[v_0, \dots, \hat{v}_k, \cdots, v_n]$ to indicate the kth $n-1$dimensional face
- The boundary of Δ, written $\partial\Delta^n$ is the union of all its proper faces
- The interior is defined to be $(\Delta^n)^\circ = \Delta^n \setminus \partial\Delta^n$

It is easy to see that any face of an n-simplex is an $n-1$ simplex in its own right.

Lemma 1.1.4 Any two k-simplexes where one in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ are homeomorphic.

1.2 Barycentric Subdivision

Barycentric subdivisions is the main ingredient for proving one of the major results of singular homology. It is defined first through Δ-set, and then passed on to singular n-simplexes, which we will see in the next chapter.

Definition 1.2.1 (Barycenter) Let $v = [v_0, \dots, v_n]$ be an n-simplex. Define the barycenter of v to be the point

$$B(v) = \frac{1}{n+1} \sum_{i=0}^n v_i$$

In fact, we can find the barycenter inductively. Knowing the barycenter b_i on the ith face $[v_0, \dots, \hat{v}_i, \dots, v_n]$, let l_i be the line connecting b_i and v_i. Then b is the intersection of all these lines l_i.

Definition 1.2.2 (Barycentric Cone) Let v be an n-simplex. Let $w = [w_1, \dots, w_n] \subseteq v$ be an $(n-1)$-simplex in v. Define the barycentric cone of w to be the n-simplex

$$\mathcal{B}(w) = [B(v), w_1, \dots, w_n] \subset v$$

Intuitively, this means that given a face of Δ^n which is an $(n-1)$-simplex, its barycentric cone is the n-simplex constructed from the vertices of the face together with the barycenter.

Definition 1.2.3 (The Barycentric Map) Let v be an n-simplex. Define the barycentric map of v to be the map

$$\mathcal{B} : C_{n-1}(v) \to C_n(v)$$

defined on the basis by $\mathcal{B}([w_1, \dots, w_n]) = [B(v), w_1, \dots, w_n]$.

Definition 1.2.4 (Barycentric Subdivision) Let Δ^n be the standard n-simplex. Define inductively the barycentric subdivision
$$S^\Delta(\Delta^n) \in C_n(\Delta^n)$$
of Δ^n by the following recursive formula.

- When $n = 0$, $S^\Delta(\Delta^0) = \Delta^0$
- When $n > 0$, the barycentric subdivision of Δ^n is given by

$$S^\Delta(\Delta^n) = \sum_{i=0}^{n} (-1)^i \mathcal{B} S^\Delta(\partial_i \Delta^n)$$

To elicit a few examples, consider the case of $n = 1$. Then we have

$$\begin{aligned} S(\Delta^1) &= \mathcal{B}S(\partial\Delta^1) \\ &= \mathcal{B}S(\partial_0\Delta^1) - \mathcal{B}S(\partial_1\Delta^1) \\ &= \mathcal{B}[v_1] - \mathcal{B}[v_0] \\ &= [b, v_1] - [b, v_0] \end{aligned}$$

where $b = \frac{1}{2}(v_0 + v_1)$. Intuitively, we are breaking up the n-simplex Δ^n into tiny pieces of n-simplexes using the center of mass of each subsimplex in Δ^n.

Lemma 1.2.5 Let $\sigma = [w_1, \dots, w_n] \subseteq \Delta^n$ be an $(n-1)$-simplex. Then the following are true.

- $\partial(\mathcal{B}(\sigma)) + \mathcal{B}(\partial\sigma) = \sigma$
- $\partial S(\Delta^n) = S(\partial\Delta^n)$

Proof We have that

$$\begin{aligned} \partial\mathcal{B}[w_1, \dots, w_n] &= \partial[b, w_1, \dots, w_n] \\ &= \sum_{i=0}^{n} (-1)^i \partial_i [b, w_1, \dots, w_n] \\ &= [w_1, \dots, w_n] - \mathcal{B}(\partial[w_1, \dots, w_n]) \end{aligned}$$

And thus the first identity is satisfied.
We prove the second item inductively. When $n = 0$, we have 0 on both sides. When $n > 0$, we have

$$\begin{aligned} \partial S\Delta^n &= \partial\mathcal{B}(S(\partial\Delta^n)) \\ &= \mathrm{id}(S\partial\Delta^n) - \mathcal{B}(\partial(S\partial\Delta^n)) && \text{(First Identity)} \\ &= S\partial\Delta^n - \mathcal{B}(S\partial^2\Delta^n) && \text{(Induction)} \\ &= S\partial\Delta^n \end{aligned}$$

■

Lemma 1.2.6 Let $v = [v_0, \dots, v_n]$ be a simplex. Let $[w_0, \dots, w_n]$ be a component in the barycentric subdivision of v. Then we have

$$\mathrm{diam}([w_0, \dots, w_n]) \leq \frac{n}{n+1} \mathrm{diam}([v_0, \dots, v_n])$$

Proof If $n = 0$, then the claim is true since $[w_0] = [v_0]$ has diameter 0. Assume that $n > 0$. We first prove the following subclaim: For every $x \in [x_0, \dots, x_n]$ an n-simplex, its maximum distance to points in the simplex is attained at a vertex x_i. Indeed if $y \in [x_0, \dots, x_n]$ with $\|x - y\|$ maximal, then we can write $y = \sum_{i=1}^{n} t_i x_i$ with $\sum_{i=1}^{n} t_i = 1$ and $t_i \geq 0$. Then we have that

$$\begin{aligned}\|x - y\| &= \left\| x - \sum_{i=1}^{n} t_i x_i \right\| \\ &= \left\| \sum_{i=1}^{n} t_i (x - x_i) \right\| \\ &\leq \sum_{i=1}^{n} t_i \|x - x_i\| \\ &\leq \max \|x - x - i\|\end{aligned}$$

with equality if y is one of the vertices with $\|x - x_i\|$ being maximal. By applying the above twice, we have that the diameter of $[w_0, \dots, w_n]$ is the length of the longest edge $[x, y]$ in $[w_0, \dots, w_n]$. Now there are two cases.

Case 1: None of x, y is the barycenter b of $[w_0, \dots, w_n]$. Then they must be the vertices of a simplex in the barycentric subdivision of one of the faces $[v_0, \dots, \hat{v}_i, \dots, v_n]$. By induction, we have that

$$\begin{aligned}\text{diam}([w_0, \dots, w_n]) &= \|x - y\| \\ &\leq \frac{n-1}{n} \text{diam}([v_0, \dots, \hat{v}_i, \dots, v_n]) \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n])\end{aligned}$$

and so we are done.

Case 2: Without loss of generality, $x = b$.
Then y lies on some face of $[v_0, \dots, v_n]$ and the claim above implies that we can take $y = v_i$ for some vertex v_i of that face. Let b_i be the barycenter of $[v_0, \dots, \hat{v}_i, \dots, v_n]$. Then we have that the barycenter of $[v_0, \dots, v_n]$ is

$$b = \frac{1}{n+1} \sum_{j=1}^{n} v_j = \frac{1}{n+1} v_i + \frac{n}{n+1} b_i$$

We thus have that

$$\begin{aligned}\text{diam}([w_0, \dots, w_n]) &= \|v_i - b\| \\ &\leq \frac{n}{n+1} \|v_i - b_i\| \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n])\end{aligned}$$

and so we conclude. ∎

2 Simplicial and Delta Complexes

2.1 Delta Complexes

Definition 2.1.1 (Δ-Complexes)
A Δ-complex $(S_\bullet, d_\bullet)$ consists of the following data.
- For each $n \in \mathbb{N}$, a set S_n of n-simplicies.
- For each $n \in \mathbb{N}$ and $0 \leq i \leq n$, a map $d_i^n : S_n \to S_{n-1}$ such that

$$d_i^{n-1} \circ d_j^n = d_{j-1}^{n-1} \circ d_i^n$$

called the face relation is satisfied whenever $i < j$.

In particular, d_i sends an n-simplex to its ith face, which is an $n-1$ simplex. This means that for $s = [v_1, \dots, v_n] \in S_n$, $d_i(s) = [v_1, \dots, \hat{v}_i, \cdots, v_n]$.

Definition 2.1.2 (Geometric Realization of Δ-Complexes)
Let $S = (S_\bullet, d_\bullet)$ be a Δ-set. The geometric realization of S is a topological space X that is built up inductively as follows:
- The 0-skeleton X^0 is a discrete set with points in S_0
- Given the $n-1$ skeleton X^{n-1} and S_{n-1}. Now define

$$X^n = \left(X^{n-1} \cup \coprod_{\alpha \in I_n} \Delta_\alpha^n \right) / \sim$$

where $\sim$ is the equivalence relation $\Delta_\beta^{n-1} \sim \partial_k \Delta_\alpha^n$ given from the face maps $d_\bullet$. (Intuitively, each face of the n-simplex in S_n gets identified with a $n-1$ simplex in X^{n-1})
- Define $X = \bigcup_n X^n$. The minimal n for which $X = X^n$ is called the dimension of n

We also write X as $|S|$ to indicate the Δ-set.

Definition 2.1.3 (Map of Delta-Complexes) Let $S = (S_\bullet, d_\bullet)$ and $S' = (S'_\bullet, d'_\bullet)$ be two Δ-sets. A map of Δ-sets $f : S \to S'$ is a family of maps $f_n : S_n \to S'_n$ for each $n \in \mathbb{N}$ such that for all $0 \leq i \leq n$, we have that

$$d'_i \circ f_n = f_{n-1} \circ d_i : S_n \to S'_{n-1}$$

2.2 Simplicial Complexes

Definition 2.2.1 (Simplicial Complexes) Let $\mathcal{K}$ be a set of simplexes. We say that $\mathcal{K}$ is a simplicial complex if the following are true.
- For any $S \in \mathcal{K}$, every face $\partial_k S$ of S lies in $\mathcal{K}$.
- If S_1 and S_2 are two simplexes in $\mathcal{K}$, then either $S_1 \cap S_2$ is empty or $S_1 \cap S_2$ is a common face of both S_1 and S_2.

Theorem 2.2.2 Every simplicial complex is a Δ-complex.

Definition 2.2.3 (Simplicial Maps)
Let $\mathcal{K}, m\mathcal{L}$ be simplicial complexes. Let $f : \mathcal{K} \to \mathcal{L}$ be a function. We say that f is a simplicial map if the following are true.
- There exists a simplex $T \in \mathcal{L}$ such that $f(S) \subseteq T$.
- For all $S \in \mathcal{K}$, $f|_S$ is affine linear.

Definition 2.2.4 (Piecewise Linear Maps)
Let $\mathcal{K}, m\mathcal{L}$ be simplicial complexes. Let $f : \mathcal{K} \to \mathcal{L}$ be a function. We say that f is piecewise linear if there is a subcomplex $\mathcal{K}' \subseteq \mathcal{K}$ such that $f|_{\mathcal{K}'}$ is a simplicial map.

2.3 Abstract Simplicial Complexes

Definition 2.3.1 (Abstract Simplicial Complex)
Let T be a set. Let $S \subseteq \mathcal{P}(T)$. We say that S is an abstract simplicial complex if for all $U \in S$ and $V \subseteq U$, we have $V \in S$.

We mostly consider the case where T is finite. In this case, we may replace the underlying set T by $\{1, \dots, n\}$ by a suitable $n \in \mathbb{N}$.

Example 2.3.2
The set $S = \mathcal{P}(\{1, 2, 3\}) \cup \mathcal{P}(\{1, 3, 4\}) \cup \mathcal{P}(\{2, 4\})$ is an abstract simplicial complex.

Definition 2.3.3 (Geometric Realization of Abstract Simplicial Complexes)
Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Define the geometric realization of S to be the space

$$|S| = \bigcup_{I \in S} \Delta^I$$

where Δ^I is the $|I|$-simplex with vertices e_i for $i \in I$.

By definition, the geometric realization $|S|$ is a subcomplex of the standard n-simplex $[e_1, \dots, e_{n+1}] \subseteq \mathbb{R}^{n+1}$.

Example 2.3.4
Let $n \in \mathbb{N}$. Consider the set $S = \mathcal{P}(\{1, \dots, n\})$. Then $|S|$ is the standard n-simplex.

2.4 Cubical Embeddings of Simplicial Complexes

Definition 2.4.1 (The Cubical Embedding)
Let $n \in \mathbb{N}$. Define the cubical embedding $i_c : \Delta^n \to [0, 1]^{n+1}$ as follows.

1. For $1 \leq k \leq n$, the barycenter of the k-dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_{n+1}) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \to [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n, $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Lemma 2.4.2
Let $n \in \mathbb{N}$. Then $\partial[0, 1]^n$ is a simplicial complex.

Definition 2.4.3 (Cubical Embeddings of Abstract Simplicial Complexes)
Let $\mathcal{K}$ be an abstract simplicial complex. Define the cubical embedding of $\mathcal{K}$ to be the subspace

$$\text{cc}(\mathcal{K}) = i_c(|\mathcal{K}|)$$

3 Polytopes, Polyhedrons and Cones

3.1 Basic Definitions

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in $\mathbb{R}^n$ since Euclidean spaces maybe identified with $\mathbb{R}^n$ canonically.

Definition 3.1.1 (Polyhedrons)
A polyhedron in $\mathbb{R}^n$ is an intersection of finitely many halfspaces.

Definition 3.1.2 (Polytopes)
A polytope in $\mathbb{R}^n$ is a polyhedron that is bounded.

Definition 3.1.3 (Dimension)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dimension of P to be the dimension of the smallest affine subspace containing P.

Definition 3.1.4 (Affinely Isomorphic)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are affinely isomorphic if there exists an affine transformation $f : \mathbb{R}^{n_1} to \mathbb{R}^{n_2}$ such that $f|_{P_1}$ is a bijection.

Note that f itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

Definition 3.1.5 (Cones)
A cone in $\mathbb{R}^n$ is a subset $C \subseteq \mathbb{R}^n$ such that if $c_1, \dots, c_n \in C$, then $\sum_{i=1}^n \lambda_i c_i \in C$ for all $\lambda_1, \dots, \lambda_n \geq 0$.

Definition 3.1.6 (Conical Hull)

3.2 The Alternate Definition and its Equivalence

Theorem 3.2.1 (The Main Theorem for Polytopes)
Let $S \subseteq \mathbb{R}^n$ be a subset. Then S is a polytope if and only if there exists a finite set of points T such that the smallest convex set containing T is S.

3.3 Faces

Definition 3.3.1 (Supporting Hyperplane)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Let $H \subseteq \mathbb{R}^n$ be a hyperplane. We say that H is a supporting hyperplane for P if the following are true.
- $P \cap H \neq \emptyset$.
- P lies entirely in one of the halfspaces defined by H.

Definition 3.3.2 (Faces)
Let $P \subseteq \mathbb{R}^n$ be a polytope. A face of P is a subset of P defined by

$$F = P \cap H$$

where H is a supporting hyperplane for P.

Lemma 3.3.3
Let $P \subseteq \mathbb{R}^n$ be a polytope. Let $F \subseteq P$ be a face. Then F is a polytope.

Definition 3.3.4 (Vertices, Edges and Facets)
Let $P\subseteq\mathbb{R}^n$ be a polytope. Let $F\subseteq P$ be a face.

- We say that F is a vertex if $\dim(F)=0$.
- We say that F is an edge if $\dim(F)=1$.
- We say that F is a facet if $\dim(F)=\dim(P)-1$.

Definition 3.3.5 (The Face Poset)
Let $P\subseteq\mathbb{R}^n$ be a polytope. Define the face poset $\mathcal{F}(P)$ of P to be the set of faces of P together with inclusion.

Definition 3.3.6 (Combinatorially Equivalent)
Let $P_1\subseteq\mathbb{R}^{n_1}$ and $P_2\subseteq\mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are combinatorially equivalent if there exists a inclusion preserving bijection between $\mathcal{F}(P_1)$ and $\mathcal{F}(P_2)$. In this case we write $P_1\simeq P_2$.

4 Polar Sets

4.1 Polar Sets

Definition 4.1.1 (Polar Sets)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the polar set of S to be

$$S^* = \{f \in (\mathbb{R}^n)^* \mid f(x) \leq 1 \text{ for all } x \in S\}$$

Proposition 4.1.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. If $0 \in P$, then P^* is a polytope.

Example 4.1.3
The polar set of the cube $C^3 = \{(a, b, c) \in \mathbb{R}^3 \mid -1 \leq a, b, c \leq 1\}$ is the octahedron

$$(C^3)^* = \{(x, y, z) \in (\mathbb{R}^3)^* \mid \pm x + \pm y + \pm z \leq 1\}$$

$P = (P^*)^*$.

$\mathcal{F}(P)^{\text{op}} \cong \mathcal{F}(P^*)$.

4.2 Simple and Simplicial Polytopes

Definition 4.2.1 (Simple Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simple polytope if for all vertices $v \in P$, there exists n edges $e_1, \dots, e_n \subseteq P$ such that $v \in \partial e_i$ for all i.

Definition 4.2.2 (Simplicial Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simplicial polytope if all its facets are simplicies.

Proposition 4.2.3
Let $P \subseteq \mathbb{R}^n$. Then P is simple if and only P^* is simplicial.

Duality of d-dimensional faces with $(n - d)$-dimensional faces.

Proposition 4.2.4
Let $P \subseteq \mathbb{R}^n$ be a polytope. If P is both simple and simplicial, then P belongs to one of the following types of polytopes.

- $n = 2$ (and so P is a polygon).
- P is a simplex.

5

Pre-req: Commutative Algebra 2, Simplicial Complexes

5.1 Stanley-Reisner Rings

Theorem 5.1.1
Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\text{Abstract Simplicial Complexes on } \{1,\dots,n\} \overset{1:1}{\leftrightarrow} \text{Squarefree monomial ideals of } R[x_1,\dots,x_n]$$

given by sending a simplicial set S to the ideal $\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$.

Definition 5.1.2 (Stanley-Reisner Rings)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of S to be

$$k[S] = \frac{k[x_1,\dots,x_n]}{\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle}$$

The ideal $I_S = \left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$ is called the Stanley-Reisner ideal of S.

Example 5.1.3
The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$k[\Delta^n] = k[x_1,\dots,x_n]$$

Example 5.1.4
The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$k[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 5.1.5
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_S = \bigcap_{J\in S} \langle x_i \mid i \notin J \rangle$$

Maps of simplicial sets -> Maps of k-algebra.

5.2 The Hilbert Series of Stanley-Reisner Rings

Definition 5.2.1 (The Face Vector)
Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the face vector $(f_{-1}, f_0, \dots, f_n)$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } S$$

for $0 \le i \le n$ and by convention, $f_{-1} = 1$.

Definition 5.2.2 (The h-Vector)
Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Let $(f_0,\dots,f_n)$ be the face vector of S.

Define the h-vector $(h_0, \ldots, h_n)$ of S by the formula

$$h_i = \sum_{k=0}^{i} (-1)^{i-k} \binom{n+1-k}{i-k} f_{k-1}$$

Lemma 5.2.3
Let S be an abstract simplicial complex on $\{1, \ldots, n\}$. Let $(f_0, \ldots, f_n)$ be the face vector of S. Let $(h_0, \ldots, h_n)$ be the h-vector of S. Then the following are true.

- We have
$$\sum_{k=0}^{n+1} f_{k-1}(t-1)^{d-k} = \sum_{j=0}^{n+1} h_j t^{n+1-j}$$
for variable t.
- For $0 \leq i \leq n$, we have
$$f_i = \sum_{k=0}^{i+1} \binom{n+1}{i+1-k} h_k$$

Theorem 5.2.4 (Stanley)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \ldots, n\}$. Let $(f_0, \ldots, f_n)$ be the face vector of S. Let $(h_0, \ldots, h_n)$ be the h-vector of S. Then the Hilbert series of $k[S]$ is given by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1} \left(\frac{t^2}{1-t^2} \right)^k = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} h_i t^{2i}$$

Example 5.2.5 | Let Δ^n be the standard n-simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by
$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$
- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by
$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

6 Moment-Angle Complexes

Pre req: Equivariant (co)homology, homological algebra, PL manifolds?

6.1 Basic Definitions

Proposition 6.1.1
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Suppose that $(S^1)^n \subseteq \mathbb{C}^n$ act on $\mathbb{C}^n$ by left multiplication. Then μ descends to a homeomorphism of orbit spaces

$$\frac{\mathbb{C}^n}{(S^1)^n} \cong \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i\} \subseteq \mathbb{R}^n$$

Lemma 6.1.2
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Then the above homeomorphism induces a homeomorphism

$$\frac{D^n}{(S^1)^n} \cong [0, 1]^n$$

Recall the construction of the cubical embedding $i_c : \Delta^n \to [0, 1]^{n+1}$.

1. For $1 \leq k \leq n$, the barycenter of the k-dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_n) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \to [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n, $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Since every finite abstract simplicial complex can be realized as a subcomplex of Δ^n for large n, we also get the cubical embedding of a finite abstract simplicial complex, which is denoted by $\text{cc}(\mathbb{K})$.

Definition 6.1.3 (Moment-Angle Complex)
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Define the moment-angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = \mu^{-1}(\text{cc}(\mathcal{K})) \subseteq D^n$$

Equivalently, it is the pullback of the inclusion map $\mathcal{Z}_{\mathcal{K}} = \text{cc}(\mathcal{K}) \to [0, 1]^n$ and the quotient map $\mu : D^n \to [0, 1]^n$.

Example 6.1.4
Let $n \in \mathbb{N}$. Then the following are true.
- $\mathcal{Z}_{\emptyset} = (S^1)^n$.
- $\mathcal{Z}_{\Delta^n} = D^n$.

Lemma 6.1.5
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. The following are true regarding the moment-angle complex of $\mathcal{K}$.
- $\mathcal{Z}_{\mathcal{K}}$ is a $(S^1)^n$-invariant subspace.
- The homeomorphism $\frac{D^n}{(S^1)^n} \cong [0, 1]^n$ induced by the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$ descends to a homeomorphism
$$\frac{\mathcal{Z}_{\mathcal{K}}}{(S^1)^n} \cong \text{cc}(\mathcal{K})$$

Recall that $\text{cc}(\mathcal{K})$ is equal to the union of a finite number of faces of $[0, 1]^n$. Also recall the notation

$$C_I = \{(x_1, \dots, x_n) \in [0, 1]^n \mid x_i = 0 \text{ if } i \in I\}$$

for a face of the cube $[0, 1]^n$ that contains the origin and $I \subseteq \{1, \dots, n\}$.

Lemma 6.1.6

Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Suppose that $\text{cc}(\mathcal{K}) = \bigcup_{I \in \mathcal{K}} C_I$. Then there is a homeomorphism

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \mu^{-1}(C_I) = \bigcup_{I \in \mathcal{K}} \{(x_1, \dots, x_n) \in D^n \mid |z_j|^2 = 1 \text{ for all } j \in \{1, \dots, n\} \setminus I\}$$

where μ is the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$.